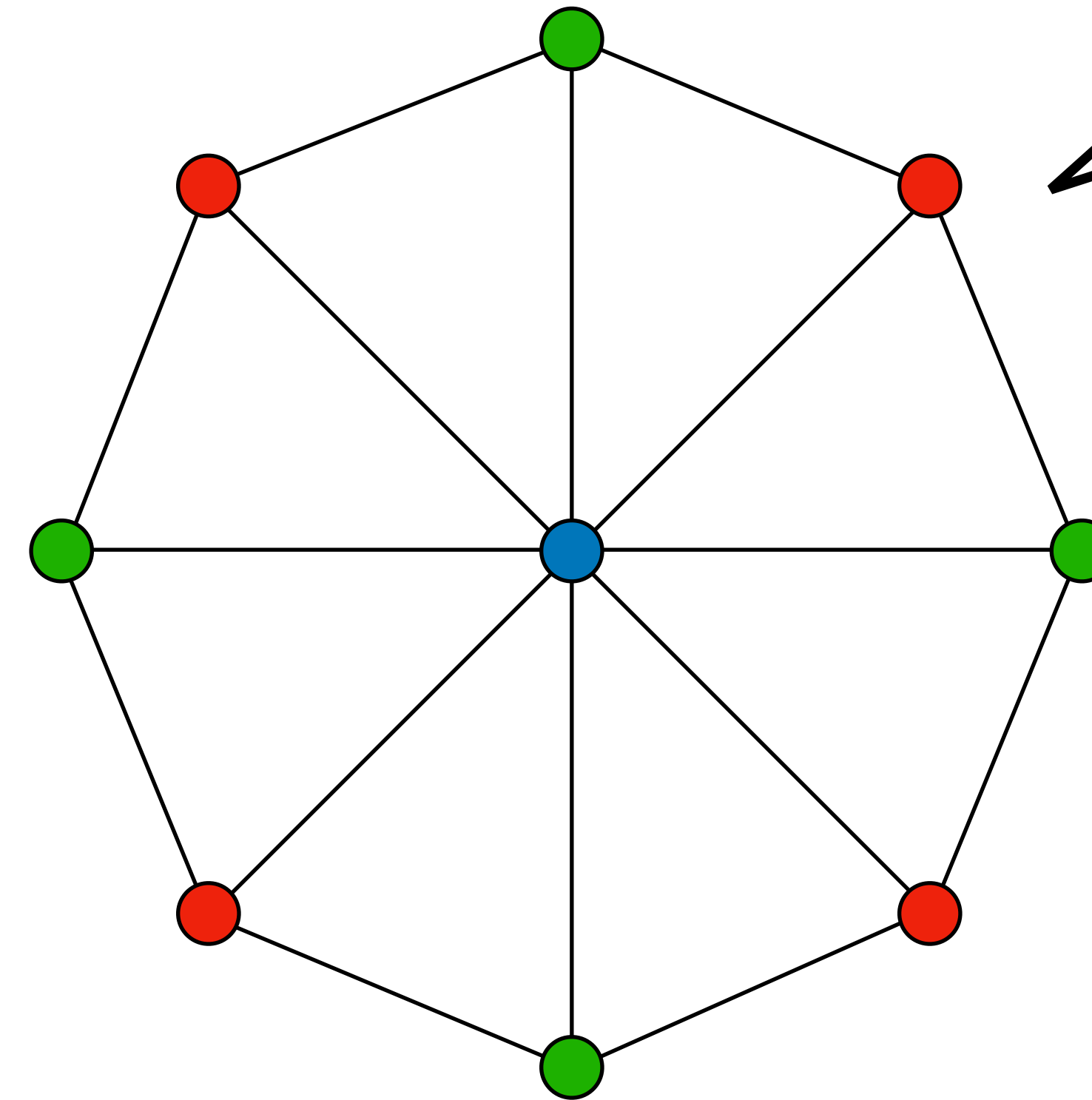
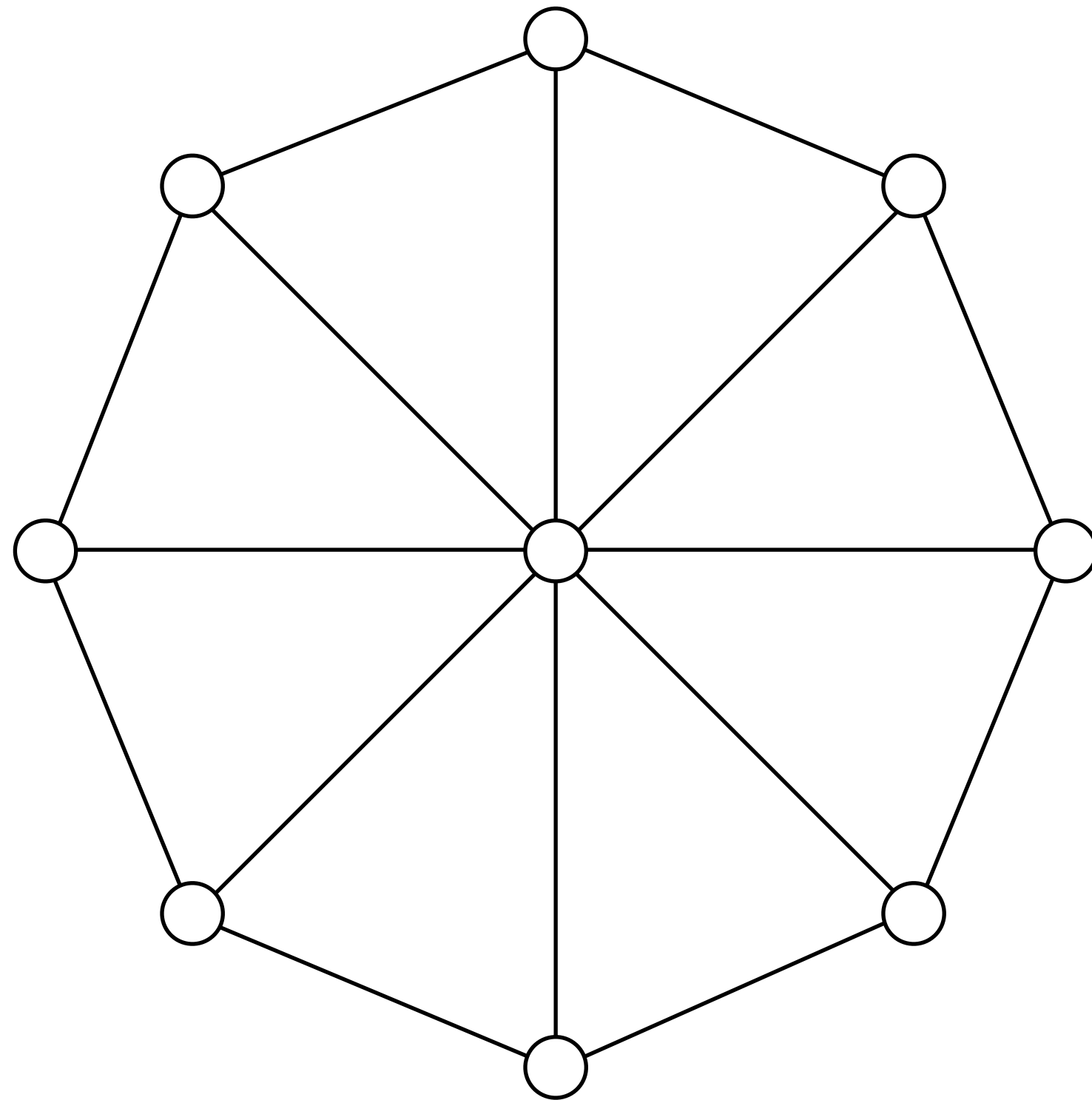


Coloring 3-Colorable Graphs with Low Threshold Rank

Jun-Ting (Tim) Hsieh, MIT

SODA 2026

Graph Coloring



3-colorable

Computation task: Given a 3-colorable graph, find a coloring that uses as few colors as possible.

3-Colorable Graphs

Color 3-colorable graphs (in polynomial-time) [Wigderson'83, Blum'94, Karger-Motwani-Sudan'98, ...].

- Trivial: n colors.
- Best algorithm: $n^{0.197}$ colors [Kawarabayashi-Thorup-Yoneda'24].

Hardness: Unique-Games*-hard (UG) to find any independent set of size ϵn [Dinur-Mossel-Regev'06].

- Cannot color with $O(1)$ colors.

Worst case graphs are very hard!

Additional Structures

Do nontrivial algorithms exist under additional assumptions on the input graphs?

k -colorable **random** graphs:

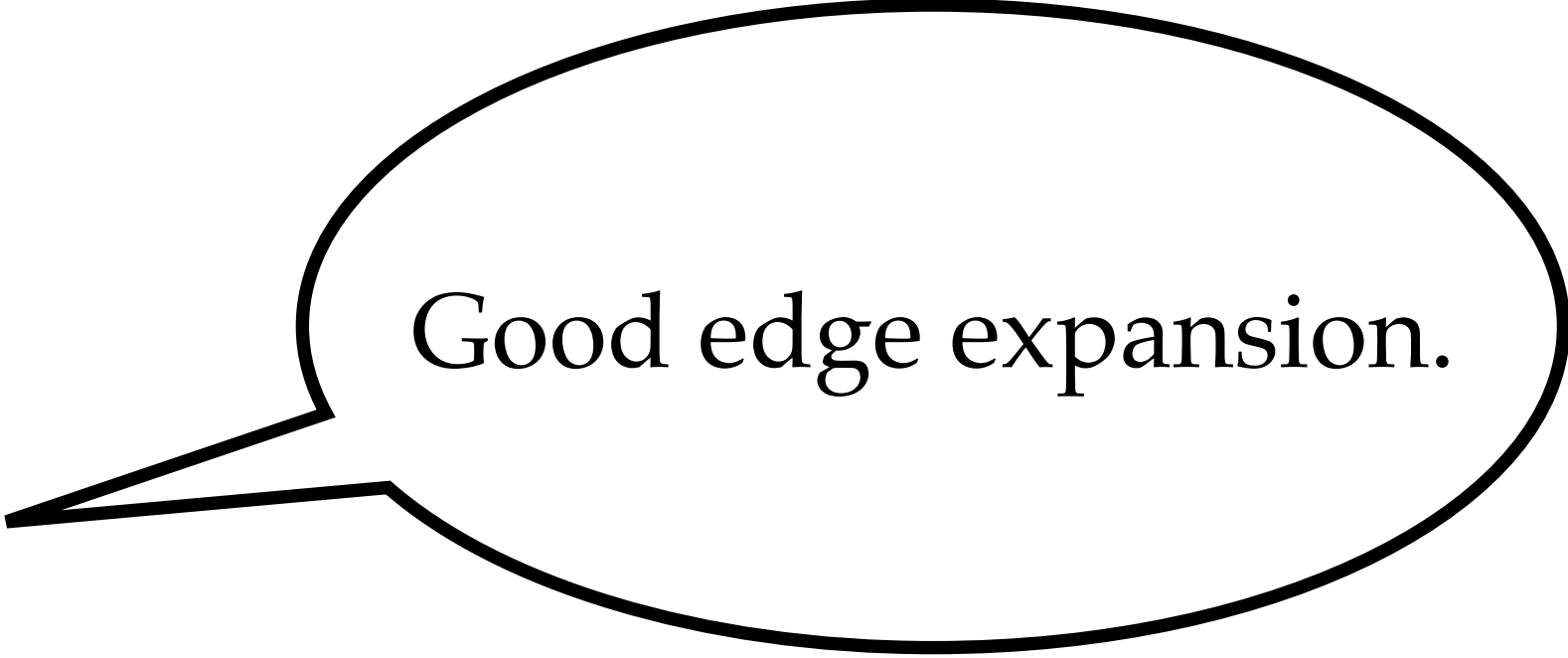
- Bottom eigenvectors $\implies$ info about color classes [Blum-Spencer'95, Alon-Kahale'97].

“Planted” k -coloring on **2-sided expanders** — $\max(\lambda_2(A), |\lambda_n(A)|) \leq \delta$:

- Bottom eigenspace enumeration [Arora-Ge'11, David-Feige'16].

One-Sided Expanders

k -colorable **1-sided expanders** — $\lambda_2(A) \leq \delta$.



Good edge expansion.

[Bafna-H.-Kothari STOC'25]:

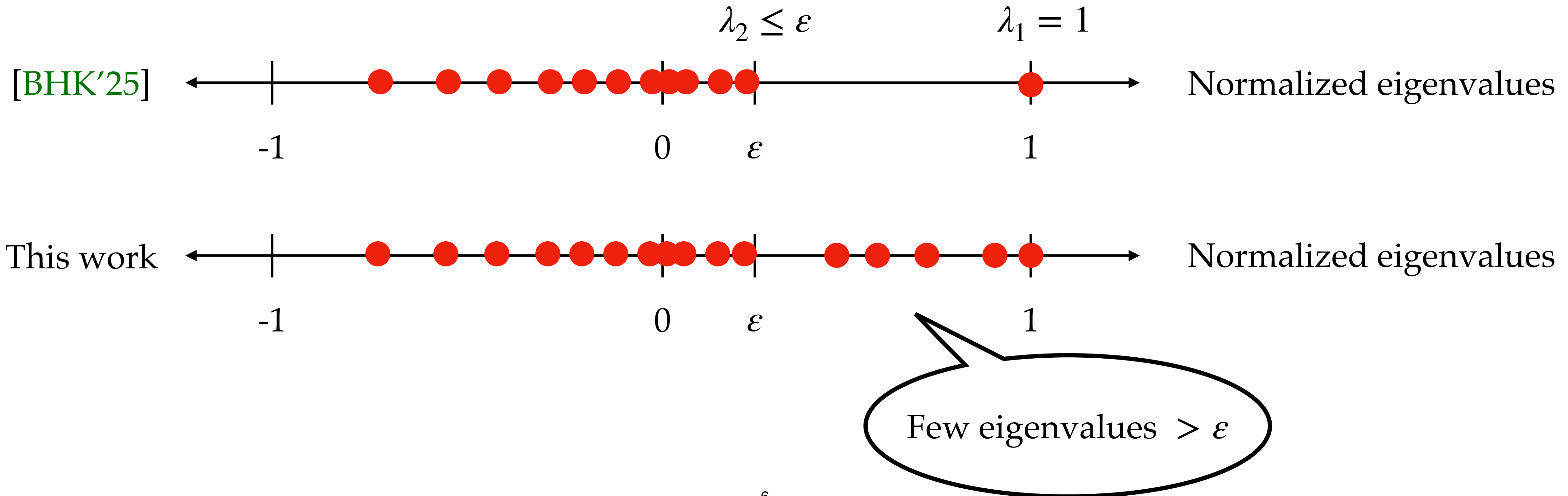
- **Algorithm:** Given an almost **3-colorable** graph with $\lambda_2 \leq 10^{-4}$, there is a polynomial-time algorithm that finds an independent set of size $\Omega(n)$.
 - Note: a color class forms an independent set.
- **Hardness:** For any constant $\varepsilon > 0$, it is UG-hard to find an independent set of size εn in an almost **4-colorable** graph with $\lambda_2 \leq o_n(1)$.

Fundamental difference between 3 and 4!

Our Results

1-sided expanders $\longrightarrow$ **low threshold rank** graphs.

Better guarantees, much simpler analysis.



Our Results

Theorem 1. Given an n -vertex 3-colorable graph with $\leq r$ eigenvalues larger than $\varepsilon > 0$, there is an algorithm that finds a proper 3-coloring on $\left(\frac{1}{2} - O(\varepsilon)\right)n$ vertices in time $n^{O(r/\varepsilon^2)}$.

◦ Better guarantee: This implies finding an independent set of size at least $\left(\frac{1}{6} - O(\varepsilon)\right)n$.

Hardness: UG-hard for 4-colorable graphs [Bafna-H.-Kothari'25].

Concurrent work [Buhai-Hua-Steurer-VariKakas'25]. UG-hard to color $> \frac{1}{2}$ fraction of vertices with $O(1)$ colors (even assuming $\lambda_2 \leq \varepsilon$).

Our Results

Our techniques extend to the independent set problem (same as [Bafna-H.-Kothari'25]).

Theorem 2. Given an n -vertex graph that contains an independent set of size $\left(\frac{1}{2} - \delta\right)n$ and has $\leq r$ eigenvalues larger than $\varepsilon > 0$, there is an algorithm that finds an independent set of size $\left(\frac{1}{2} - \text{poly}(\delta, \varepsilon)\right)n$ in time $n^{r/\text{poly}(\varepsilon)}$.

- [Bafna-H.-Kothari'25] only finds one of size $\Omega(n)$, assuming small λ_2 .

Subexponential-Time Algorithm?

Open question: Is there a **subexponential-time** algorithm that colors a 3-colorable graph with $n^{o(1)}$ colors?

- Recall: Best polynomial-time algorithm uses $n^{0.197}$ colors.

Success story of **Unique Games**:

- Algorithms when the graph is an expander [Trevisan'08, AKKSTV'08].
- Algorithms when the graph has low threshold rank [Kolla'11, Makarychev-Makarychev'11].
- Subexponential-time algorithm for worst-case instances [Arora-Barak-Steurer'15].

Technical Overview

Distribution over 3-Colorings

Imagine we have a **distribution** μ over proper 3-colorings (domain $x \in \{1,2,3\}^n$).

Simple observation: For a vertex u , if $\Pr_{\mu}[x_u = \sigma] > 1/2$ for some color σ (i.e., *concentrated*), then we can color u with σ .

This gives a **valid partial coloring**.

- For any edge (u, v) , $\Pr_{\mu}[x_u = x_v] = 0$.
- There cannot be any color σ such that both $\Pr_{\mu}[x_u = \sigma], \Pr_{\mu}[x_v = \sigma] > 1/2$.

Distribution over 3-Colorings

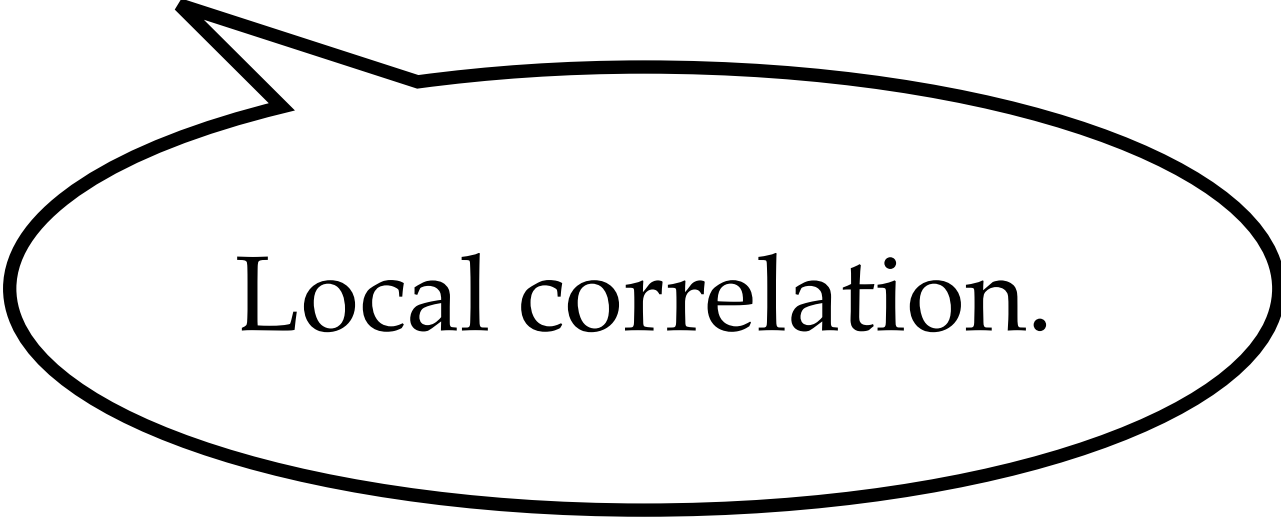
Bad set:

$$S = \left\{ u \in V(G) : \Pr_{\mu}[x_u = \sigma] \leq 1/2, \forall \sigma \in [3] \right\}$$

Ideal: $|S|$ is small.

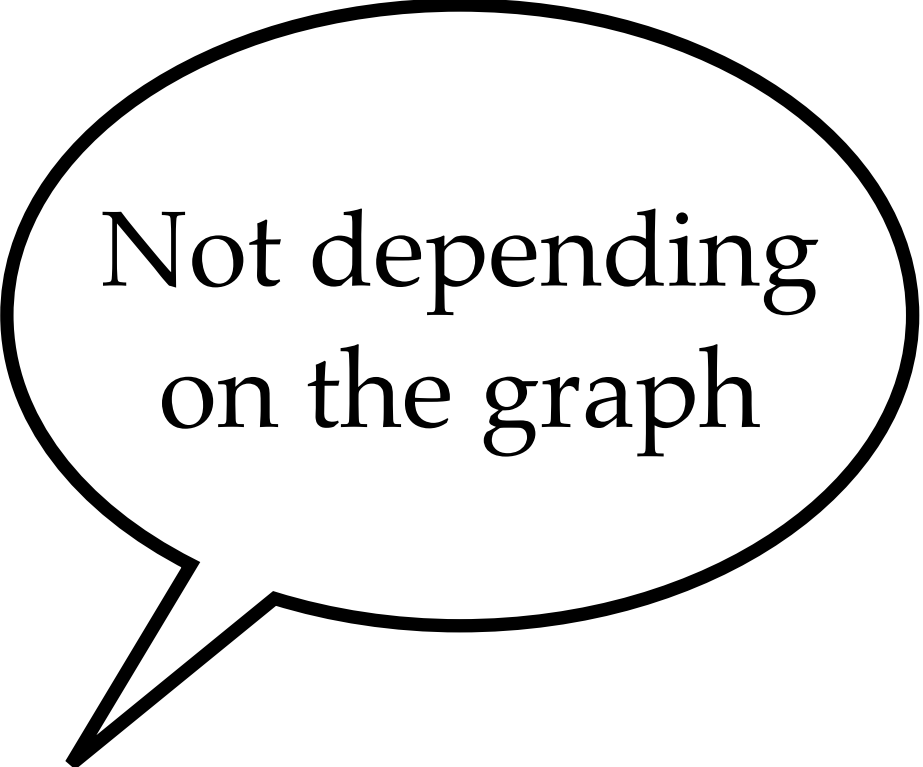
Main lemma: For any edge (u, v) with $u, v \in S$, the random variables (X_u, X_v) must have a **large “correlation”**.

If $|S| > \left(\frac{1}{2} + \varepsilon\right) n$, then the expected correlation over the edges is $\geq \Omega(\varepsilon)$.



Local correlation.

Local vs. Global Correlation



Not depending
on the graph

Standard techniques [Barak-Raghavendra-Steurer'11, Guruswami-Sinop'11]:

1. For **low threshold rank** graphs, **large local correlation** $\implies$ **large global correlation**.
2. **Global correlation rounding**: **reduce global correlation** by iterative conditioning.

Algorithm: given a distribution μ over proper 3-colorings,

1. Iteratively condition to reduce the global correlation.
2. Color the “concentrated” vertices.
3. If the bad set $|S| > (1/2 + \varepsilon)n$, then large local correlation $\implies$ large global correlation.

Pseudo-Distribution

We've assumed that we have access to a distribution μ over 3-colorings.

Degree- d **Sum-of-Squares** algorithm $\longrightarrow$ degree- d pseudo-distribution $\tilde{\mu}$.

- The marginal of $\tilde{\mu}$ on any d vertices is a true distribution.
- Runs in time $n^{O(d)}$.

For our purpose, if the threshold rank is r , then the global correlation rounding step requires $d \geq O(r/\epsilon^2)$.

$\implies$ runtime $n^{O(r/\epsilon^2)}$.

Subexponential-Time Algorithm?

Open question: Is there a **subexponential-time** algorithm that colors a 3-colorable graph with $n^{o(1)}$ colors?

- Recall: Best polynomial-time algorithm uses $n^{0.197}$ colors.

Thank you!